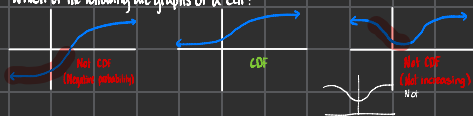


Continuous Random Variables

- Any value within an uncountably infinite number of values
- Ex. height, distance, time
- A random variable X with cdf $F(x)$ is continuous if $F(x)$ is continuous for $-\infty < x < \infty$
 - Smooth function

Example

Which of the following are graphs of a cdf?



Probability Density Function (pdf)

- If X has a cdf, the pdf $f(x)$ is given by

$$f(x) = \frac{dF(x)}{dx} = F'(x)$$

- It follows that $F(x) = \int_{-\infty}^x f(t) dt$
- Probability of a single event is zero ($P(X=x) = 0$, for all x)
- $f(x)$ is shorthand for $f_x(x)$

Quantiles and Percentiles

If $0 < p < 1$, the quantile of order p of the distribution of X is ξ_p such that $P(X < \xi_p) \leq p$ and $P(X \leq \xi_p) \geq p$.

Also known as the $100p$ percentile of X (continuous or discrete)

- $\xi_{0.5}$ is the median that $P(X < \xi_{0.5}) \leq 0.5$ and $P(X \leq \xi_{0.5}) \geq 0.5$. The median is also the second quantile q_2 .
- The difference $IQR = q_3 - q_1$ is the interquartile range of X which is used as a measure of spread or dispersion of the distribution of X
- Not required to be unique

Theorem 31

If $f(x)$ is a pdf for a continuous random variable, then

- $f(x) \geq 0$ for all $x \in \mathbb{R}$
- $\int_{-\infty}^{\infty} f(x) dx = 1$

Induced Probability Distribution

- Defined as $P_x((a, b]) = P(a \leq X \leq b) = P(\{c \in \mathbb{R} : a < X(c) \leq b\}) = \int_a^b f(x) dx$
- Area under curve from a to b
- Since $P(X=a) = P(X=b) = 0$, we have that $P(a \leq X \leq b) = P(a < X < b)$
 - Not true for discrete variables

Example

Given $f(x) = cx^2$, $0 \leq x \leq 2$, find c so that $f(x)$ is a pdf. Also find $P(1 < X < 2)$

- cx^2 is continuous for any $c \in \mathbb{R}$
- Want $\int_0^2 cx^2 dx = 1$

$$\Rightarrow c \left[\frac{x^3}{3} \right]_0^2 = c \left(\frac{8}{3} \right)$$

$$\Rightarrow 1 = c \left(\frac{8}{3} \right)$$

$$\Rightarrow c = \frac{3}{8}$$
- $P(1 < X < 2) = \int_1^2 \frac{3}{8} x^2 dx = \frac{7}{8}$

Example

Find the 0.2 quantile (20th percentile) of the distribution that has pdf $f(x) = 4x^3$ for $0 \leq x \leq 1$ (0 elsewhere)

$$\Rightarrow F(x) = \int_0^x 4t^3 dt = x^4$$

$\Rightarrow$ Want the point where $F(x) = 0.2$, so $P(X \leq x) = 0.2$, and $P(X > x) = 0.2$

$$\Rightarrow 0.2 = F(x)$$

$$\Rightarrow 0.2 = x^4 \text{ (has solutions } \pm 0.2^{1/4})$$

$$\Rightarrow P(-0.2^{1/4} \leq X \leq 0.2^{1/4}) = P(X \leq 0.2^{1/4}) \text{ (Since the range } [-1, 0] \text{ has 0 probability, we can exclude it)}$$

$$\Rightarrow \text{Therefore } q_{0.2} = 0.2^{1/4}$$